An Analysis of the Inflation Reduction Act

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Abstract

Since the Decade of the Environment, little has been done on a legislative level to address the climate crisis until Congress in late 2022 responded with a significant investment in the Inflation Reduction Act. This investment is the most critical piece of federal climate legislation ever enacted in the United States, providing $370 billion over ten years to respond to the climate emergency.[1] Analyses show that this legislation could help the US cut emissions to approximately 40% below 2005 levels by 2030.[2] Throughout the political process, the Inflation Reduction Act faced many barriers to its passage, but partisanship allowed for its survival. However, while the act's main components are to drive green development its impacts expand further than climate needs. Those impacts include: job creation, incentivizing the development of domestic industry, and decreasing the cost of living. All of these are being subsidized into existence without a damper on the federal budget. This article additionally highlights the potential impact of the Inflation Reduction Act (IRA) on equality. The act includes many additional tax credits to incentivize development in low-income communities or energy communities. While the IRA promotes worthwhile affordability and environmental goals, targeted reforms addressing limitations are still required to spread financial relief broadly across all demographics. Continued tracking of economic impacts on diverse groups and the environment is also essential to fully understand and adjust the IRA's programs over time.

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[1] Lashof, Dan, Devashree Saha, Melissa Barbanell, Jennifer Chen, Christina DeConcini, Janet Ranganathan, and Angela Anderson. 2022. "After Inflation Reduction Act: 5 Ways to Hit the US Emissions Goal." Www.wri.org, August. https://www.wri.org/insights/inflation-reduction-act-emissions-gap.

[2] princeton.edu. n.d. "REPEAT | Rapid Energy Policy Evaluation & Analysis Toolkit." Repeatproject.org. https://repeatproject.org/.

Introduction

The United States is the home of traditional environmentalism, a grassroots movement built on nearly a century of efforts to address water, air, and land contamination caused by industrialization and urbanization, culminating in the first Earth Day in 1970.[3] Progressive-era reformers in the early 1900s warned of unregulated economic development from industrialization, leading to public health crises and environmental degradation. Conservation groups mobilized early to protect wilderness areas and create the first national parks.

Accelerated activism in the 1950s and 60s developed from unprecedented affluence in post-WWII America, allowing white middle-class families to move to suburbia in search of the quality of lives far from pollution, as well as Rachel Carson's Silent Spring shocking readers of the harms of chemicals on the human body.[4] Public pressure and grassroots movements put environmentalism at the forefront of liberal politics in the 1960s, causing Presidents Kennedy, Johnson, and Nixon to pass sweeping command-and-control legislation into the 1970s.[5]

Reagan's deregulation administration and the growing chokehold of the fossil fuel industry on the Republican party from campaign contributions stalemated the environmental movement on the federal level.[6] Growing partisanship and politicizing the ecological movement, especially the placement of climate change in the environmental box, stunted legislative progress on bipartisan efforts.

Traditional environmentalism is rooted in elitism and homogeneity, started by white middle-class families in suburban areas. By alienating the disproportionately affected minorities, lower income, and the working class through preservationism policy, the environmental movement failed to address a necessary support base. With focuses on NIMBYism, anti-globalization, and conservation, traditional environmentalism is not adapted to modern issues and opposition.[7]

In a desert of climate legislation since,[8] the Inflation Reduction Act, signed into law in 2022 by President Biden, is a landmark piece of legislation that propels the United States toward a sustainable and economically competitive future—enacted during a critical juncture in economic history, where the American populous was plagued by unemployment and inflation induced by COVID-19 induced supply chain disruptions. This investment is the most critical piece of federal climate legislation ever enacted in the United States,[9] providing $370 billion over ten years to respond to the climate emergency. Analyses show that this legislation could help the US cut emissions to approximately 40% below 2005 by 2030[10]. As we navigate through the political process and impacts of the Inflation Reduction Act, we uncover nuanced legislation that emerged from a traditional environmental movement and climate necessity.

Legislative Process

Partisanship and Congressional Structures

Partisanship, the firm adherence to a particular party or cause in the legislative process, protected the Inflation Reduction Act and what it would achieve. The United States legislative body, Congress, has experienced a sharp increase in partisan behavior and polarization from the

[3] "I. Origins of the Environmental Movement · Exhibit · Give Earth a Chance: Environmental Activism in Michigan." 2018. Umich.edu. 2018. https://michiganintheworld.history.lsa.umich.edu/environmentalism/exhibits/show/main_exhibit/origins.
[4] Shutkin, William. 2021. "How American Environmentalism Failed." The MIT Press Reader. August 31, 2021. https://thereader.mitpress.mit.edu/how-american-environmentalism-failed/.
[5] Shutkin, William. 2021. "How American Environmentalism Failed." The MIT Press Reader. August 31, 2021. https://thereader.mitpress.mit.edu/how-american-environmentalism-failed/.
[6] Geary, Daniel. 2012. "Environmental Movement | Encyclopedia.com." Encyclopedia.com. 2012. https://www.encyclopedia.com/earth-and-environment/ecology-and-environmentalism/environmental-studies/environmental-movement.
[7] Lavelle, Marianne. 2016. "Fossil Fuel Money to GOP Grows, and so Does Climate Divide." Inside Climate News. September 15, 2016.
[8] Center for Climate and Energy Solutions. 2017. "Congress Climate History." Center for Climate and Energy Solutions. August 19, 2017. https://www.c2es.org/content/congress-climate-history/.
[9] Lashof, Dan, Devashree Saha, Melissa Barbanell, Jennifer Chen, Christina DeConcini, Janet Ranganathan, and Angela Anderson. 2022. "After Inflation Reduction Act: 5 Ways to Hit the US Emissions Goal." Www.wri.org, August.
[10] princeton.edu. n.d. "REPEAT | Rapid Energy Policy Evaluation & Analysis Toolkit." Repeatproject.org. https://repeatproject.org/.

2000s onward.[11] This behavior manifests in more congresspeople voting along party lines, rising allegiance of a party to their President's agenda, and fewer bipartisan bills getting passed.

Considering the state of Congress, the survival of the Inflation Reduction Act was largely dependent on Democratic leadership in the legislature and executive branches. Along with a Democratic president, the House of Representatives had 222 Democrats (4 Delegates), a majority over the 215 Republicans (1 Delegate and 1 Resident Commissioner of Puerto Rico, with four vacant seats.[12] This Democratic majority gave way for a Democrat speaker of the House, Nancy Pelosi, to use that platform to protect Democrat-sponsored legislation.

The Senate simultaneously had 50 Republicans, 47 Democrats, and 3 Independents who caucused with the Democrats.[13] With the Vice President's ability to act as the President of the Senate, V.P. Kamala Harris served as a tiebreaker vote to give Democrats a majority in both houses of Congress. Democratic leadership in the executive branch, along with slim majorities in both houses of Congress, allowed for the endurance of the Inflation Reduction Act as it went through the legislative process, ultimately passing into law without a single Republican vote.[14] However, where did it start?

Biden's Build Back Better Plan

The Inflation Reduction Act went through many phases in the legislative process, beginning with President Biden's Build Back Better plan.[15] On the White House Website, there is a loose framework of policy goals outlined by the Biden Administration, in which there is a heavy focus on climate-related investments that will "rebuild the backbone of the country – the middle class."

Specifically, the framework includes consumer rebates and tax credits to reduce costs for middle-class families electrifying and shifting to clean energy and helping rural communities tap into clean energy opportunities through targeted grants and loans through the Dept of Agriculture. The Biden Administration also aims to target incentives to grow U.S. supply chains in clean energy technology like photovoltaic panels. They plan to use policy to bolster nature's resilience to climate change through investments in coastal restoration, forest management, and soil conservation, among many others.

This White House policy framework claims to set the U.S. on course to meet its climate goals while creating millions of well-paying jobs to grow the economy and support the American Labor force. Especially when the economy is strained by inflation and high interest rates, lumping climate goals with economic growth would garner more support than purely creating mandatory and legally coercive regulatory frameworks or efficiency standards separate from financial issues. These loose frameworks outlined on President Biden's website drove policy goals for the Democratic party, almost characterizing and setting their vision for policy under his administration, citing the Build Back Better plan for climate-related bills and daily records.

Daily Records Before the Introduction of the IRA

Even before a comprehensive bill addressing President Biden's Build Back Better plan was introduced, congressional daily records mention implementing aspects of this framework. For example, on Aug. 23rd, 2021, (D) Rep. Auchincloss of Massachusetts 4th district made a

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[11] "Political Polarization in the American Public." Pew Research Center - U.S. Politics & Policy. June 12, 2014. https://www.pewresearch.org/politics/2014/06/12/political-polarization-in-the-american-public.
[12] Membership of the 117th Congress: A Profile. Congressional Research Service, 14 Dec. 2022.
[13] Membership of the 117th Congress: A Profile. Congressional Research Service, 14 Dec. 2022.
[14] "Actions - H.R.5376 - 117th Congress (2021-2022): Build Back Better Act." Www.congress.gov. November 19, 2021. https://www.congress.gov/bill/117th-congress/house-bill/5376/actions.
[15] "The Build Back Better Framework." n.d. The White House. https://www.whitehouse.gov/build-back-better.

brief statement about investing in climate resiliency to "take steps necessary to end climate change," as his district faced "a tropical storm… with intense winds of at least 47 miles per hour".[16] He combined regional concerns of intense weather-induced damage with his party's goals to advance this Build Back Better plan through various legislations.

The Daily Record of Aug. 24th, 2021, provides more information on this: where they debated three significant legislations, two of which are the Bipartisan Infrastructure Act and the adoption of S. Con. Res. 14[17], the Senate reconciliation for the budget of the 2022 fiscal year beginning on Oct. 1st, 2021. Notable arguments from democrats include Rep. Scanlon of Pennsylvania in the Rules Committee's argument that the Bipartisan Infrastructure Bill and S. Con. Res 14 are critical for creating jobs and sustainability while also revising the budget to meet Biden's Build Back Better Plan.[18]

(R) Rep. Fischbach refuted the bill's claims above benefitting all citizens, saying that "with tens of billions for electric vehicle plug-ins, Amtrak, and light rail if you live in a deep blue city, this bill is for you. However, this bill leaves you behind if you are among millions of Americans in a more rural area."[19] She claimed that this bill exists only for people in deep blue cities, where more democrats live, evidencing the perceived partisanship on the democrat-sponsored bill. Contrary to Rep. Fischbach's argument, Sections 22001 through 22005 in the IRA have explicit provisions for rural communities, directly involving the USDA Rural Development's Rural Utilities Service and Rural Business and Cooperative Service.

Furthermore, (R) Rep. Smith from Missouri argued that Democrats put Bernie's budget with a transportation bill and a voting rights bill (adding amendments) because they cannot pass Bernie's budget alone. He quoted that the country was facing a "Biden border crisis, Biden energy crisis, Biden Afghanistan crisis." [20] He makes these points to argue the increased spending under his Build Back Better plan by citing common grievances with the Biden administration. (R) Rep. McCarthy attacked Biden's failure with Afghanistan and the Taliban takeover.[21] This record was arranged for the Bipartisan Infrastructure Bill, which was eventually signed into law, to be passed in conjunction with this budget (S. Con. Res 14 that would address Build Back Better provisions, but S. Con. Res 14 failed).[22]

The Build Back Better Act

The initial version of the Inflation Reduction Act (Public Law 117-169), the Build Back Better Act (HR 5376), was sponsored by (D). Rep. Yarmuth on Sept. 27th, 2021, almost a year before its signage into law. With the nature of this bill having to do with appropriations (federal funds), it had to start in the House of Representatives.[23]

From there, the House Committee on the Budget was assigned by the Speaker of the House bill to develop, from which the committee got reports from twelve other committees related to the bill's contents.[24] A deep dive into the report generated by the Committee on Agriculture reveals partisanship in the debates and germane amendments offered regarding forestry and agriculture.

Some notable Republican-sponsored amendments include (R) Rep. Allen, who offered an amendment that would have removed funding for climate change research (failed by a vote of 24

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[16] Jacob D. Auchincloss (MA). Congressional Record Vol. 167 (2021) no. 149. - Daily Edition (Text from: Congressional Record Permanent Digital Collection). https://www.congress.gov/congressional-record/volume-167/issue-149/house-section/article/H4348
[17] "Infrastructure Investment and Jobs Act (H.R. 3684)" Congressional Record (2021). (Text from: Congressional Record Permanent Digital Collection). https://www.congress.gov/bill/117th-congress/house-bill/3684/actions
[18] Mary G. Scanlon (PA). Congressional Record Vol. 167 (2021) no. 150. - Daily Edition (Text from: Congressional Record Permanent Digital Collection). https://www.congress.gov/congressional-record/volume-167/issue-150/house-section/article/H4359
[19] Michelle Fischbach (MN). Congressional Record Vol. 167 (2021) no. 150. - Daily Edition (Text from: Congressional Record Permanent Digital Collection). https://www.congress.gov/congressional-record/volume-167/issue-150/house-section/article/H4359
[20] Jason T. Smith (MO). Congressional Record Vol. 167 (2021) no. 150. - Daily Edition (Text from: Congressional Record Permanent Digital Collection). https://www.congress.gov/congressional-record/volume-167/issue-150/house-section/article/H4359
[21] Kevin O. McCarthy (CA). Congressional Record Vol. 167 (2021) no. 150. - Daily Edition (Text from: Congressional Record Permanent Digital Collection). https://www.congress.gov/congressional-record/volume-167/issue-150/house-section/article/H4359
[22] S. Con. Res. 14" Congressional Record (2021). (Text from: Congressional Record Permanent Digital Collection). https://www.congress.gov/bill/117th-congress/senate-concurrent-resolution/14/actions
[23] U.S. Constitution Article I, Section 7, clause 1
[24] "Report of the Committee on the Budget House of Representatives to Accompany H.R. 537 Together with Minority Views." Congressional Record Permanent Digital Collection, 27 Sept. 2021, pp. 1–100,

years to 27 days).[25] Additionally, (R) Rep. Fischbach offered an amendment to make changes and provide funding for specific rural broadband programs (failed by 24 yeas to 27 nays).[26] (R). Rep. Desjarlais offered an amendment that would have reduced total spending under the title by 2/3rd (failed 24 yeas to 27 nays).[27] Apart from one (R), Rep. Fischback offered a few amendments that were successfully implemented that would insert an opposition to tax burdens on the people (27 yeas to 24 nays). The final majority views summarize how this bill will benefit Biden's Build Back Better plan, the middle class, and the economy.

Minority views provided a different story – they argued that government spending increases for COVID-19 relief have already strained the economy and that this would similarly increase deficit spending. They also criticized that Democrats wrote the legislation behind closed doors and that Republicans did not see the language until less than 24 hours before the meeting commenced (Report of the Committee on the Budget). This aspect of partisanship is a double-edged sword. Potentially threatening amendments to the IRA were less likely to develop since Republicans had little time to analyze the expansive legislation. Still, provisions that could negatively impact minorities could not be addressed by Republicans because they did not have time to find them within the bill. These views argued that money was spent for unnecessary purposes like the civilian climate corps and mature forest/species protections instead of wildfire prevention and at-risk acres. They also argued that the respect to the needs of rural businesses and water systems is "too little too late." This minority view was signed by the same 24 people united in their votes on amendments in the committee consideration process.

On the House of Representatives floor, two hours of debate were set aside to consider the Build Back Better Act.[28] The debate was regarded as unfinished business, and a motion to reconsider was laid for Nov. 19th, 2021, which took 14 days.[29] The bill was passed in the House on Nov. 19th without a Republican vote. It took until August of the following year for the Senate to receive this revised bill, the Inflation Reduction Act, and read it for the first time on Aug. 2nd, 2022. Since this was passed through the budget reconciliation project, a filibuster could not be used. Thus, there was no need for Republican votes.

A pathway the Republican party could use, however, was to use a legislative "hostage" strategy, withholding the passage of a bipartisan package that would strengthen domestic production, which is what Senate Minority Leader Mitch McConnell implemented.[30] This seemingly desperate attempt to control the sway of the legislation resulted in (D) Sen. Joe Manchin scuttling his party's reconciliation package with changes to prescription drug provisions, which then GOP senators saw no need to keep the legislation hostage.[31]

Within the democratic party, Sen. Sinema's opposition prevented the Build Back Better Act from being introduced (MSNBC). To get her on board, the Democratic party scrapped a provision to end the private equity tax on capital income at a lower 15% rate. Democratic leaders also tweaked the structure of a 15% corporate tax minimum and threw in drought relief funding to benefit Arizona.[32]

The Committee on Banking, Housing, and Urban Affairs held a hearing regarding this legislation on Aug. 4th; then amendments were offered. Two hundred ninety-three amendments were submitted in the Senate between Aug. 6th and 7th. However, only 27 roll call votes were called, as those amendments violated previous laws or were dismissed by Order of the Chair.

[25] Rick W. Allen (GA). Report of the Committee on the Budget House of Representatives to Accompany H.R. 537 Together with Minority Views. Committee on Agriculture. (2021) p. 12. (Text from: Congressional Record Permanent Digital Collection)
[26] Michelle Fischbach (MN). Report of the Committee on the Budget House of Representatives to Accompany H.R. 537 Together with Minority Views. Committee on Agriculture. (2021) p. 12. (Text from: Congressional Record Permanent Digital Collection)
[27] Scott E. Desjarlais (TN). Report of the Committee on the Budget House of Representatives to Accompany H.R. 537 Together with Minority Views. Committee on Agriculture. (2021) p. 12. (Text from: Congressional Record Permanent Digital Collection)
[28] "H.Res.774 - Providing for consideration of the bill (H.R. 5376) to provide for reconciliation pursuant to title II of S. Con. Res. 14; and for other purposes." Congressional Record (2021). (Text from: Congressional Record Permanent Digital Collection).
[29] "117th Congress Legislative Calendar." n.d. Ballotpedia. https://ballotpedia.org/117th_Congress_legislative_calendar.
[30] "Mitch McConnell's New Hostage Threat Is Brazen (Even for Him)." 2022. MSNBC.com. July 1, 2022. https://www.msnbc.com/rachel-maddow-show/maddowblog/mitch-mcconnells-new-hostage-threat-brazen-even-rcna36294.
[31] "What's Great (and What's Not) about Manchin's Deal with Schumer." 2022. MSNBC.com. July 28, 2022. https://www.msnbc.com/rachel-maddow-show/maddowblog/s-great-s-not-manchins-deal-schumer-rcna40403.
[32] "What It Took to Get Sinema to Support the Inflation Reduction Act." n.d. MSNBC.com. https://www.msnbc.com/rachel-maddow-show/maddowblog/took-get-sinema-support-inflation-reduction-act-rcna41692.

Notable amendments and their votes include Sen. Graham's proposal, which would have struck a tax increase resulting in higher consumer prices for gas, heating oil, and other energy sources for Americans making less than $400k per year, which failed by a 50-50 vote then was ruled out by Order of the Chair.[33] Another was Sen. Shelby's to end the President's War on Coal by approving coal leases, which was not agreed to in the Senate with a 50-50 vote.[34] Sen. Thune's amendment to remove harmful small business taxes was agreed to 57-43.[35] Finally, Sen. Warner's amendment that protected noncorporate businesses from excess losses and ended the extension of tax cuts for state and local taxes was agreed to 51-50.[36]

Fossil fuel lobbyists tend to target conservative members, which influences their votes in legislation to protect the lobbyists' interests, which include stopping the President's War on Coal. The IRA was sent back to the House of Representatives on Nov. 8th, 2023, where, after 3 hours of debate, the House agreed to the Senate amendments 220 yeas to 107 nays. The IRA was then passed on and signed by the President on August 16th.[37]

A methane fee was one of the significant "sticks" of this legislation. While not a carbon tax, an important policy goal for the democratic party and environmental movement, this methane fee provides an essential disincentive for using fossil fuels. This methane fee was developed through the bodies of Congress, beginning when introduced in the House, loosely establishing a methane fee.[38] When passed in the House, it added a provision for paying royalties for methane extracted from federal lands and specified offshore ones. What passed in the Senate a year later was the EPA's responsibility to use appropriated funds to provide financial incentives to monitor/reduce methane emissions and impose/collect a charge on methane emissions from an owner or operator of a facility that emitted more than 25000 metric tons of co2 equivalent greenhouse gasses per year.[39]

After The Inflation Reduction Act's Passage

Minority views from the House Committee on Oversight and Reform claim that "with massive spending packages like the 2021 American Rescue Plan Act (P.L. 117-2), the 2021 Infrastructure Investment and Jobs Act (P.L. 117-58), and the Inflation Reduction Act of 2022 (P.L. 117-169), the Biden Administration has caused consumers to experience historically high prices by overstimulating the economy."[40] These committee minority members supported H. Res. 1412, which would have conducted oversight over the Biden Administration's treasury, understanding how, in their words, "their reckless legislative spending packages would contribute to the historic levels of inflation Americans are experiencing" Rep. James Comer, Ranking Member of the Committee on Oversight and Reform.[41]

H. Res. 1412 was introduced and placed on the House of Representatives calendar, but no other actions were taken. The most likely cause of this was the Speaker of the House at the time, Nancy Pelosi, invoking Hastert rule[42], an unwritten rule that the Speaker of the House will not schedule a vote on bills that do not have consensus within their party. Since H. Res. 1412 was stunted after being put on a calendar, former Speaker Nancy Pelosi might have invoked the Hastert rule and effectively "killed" the bill.

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[33] Lindsey O. Graham (SC). S. Amdt. 5301 to S. Amdt. 5194 (2022) (Text from: Congressional Record Permanent Digital Collection). https://www.congress.gov/amendment/117th-congress/senate-amendment/5301

[34] Richard C. Shelby (AL). S. Amdt. 5418 to S. 5194 (2022). (Text from: Congressional Record Permanent Digital Collection). https://www.congress.gov/amendment/117th-congress/senate-amendment/5418

[35] John R. Thune (SD). S. Amdt. 5418 to S. 5194 (2022). (Text from: Congressional Record Permanent Digital Collection). https://www.congress.gov/amendment/117th-congress/senate-amendment/5418

[36] Mark R. Warner (VA). S. Amdt. 5418 to S. 5194 (2022). (Text from: Congressional Record Permanent Digital Collection). https://www.congress.gov/amendment/117th-congress/senate-amendment/5418

[37] "Inflation Reduction Act of 2022 (H.R. 5376)". Congressional Record (2021-2022). (Text from: Congressional Record Permanent Digital Collection).

[38] "Inflation Reduction Act of 2022 (H.R. 5376)". Congressional Record (2021-2022). (Text from: Congressional Record Permanent Digital Collection).

[39] Hansel, Peter. 2023 Interview by Hannah Midha.

[40] RESOLUTION of INQUIRY DIRECTING the SECRETARY of the TREASURY to TRANSMIT CERTAIN DOCUMENTS to the HOUSE of REPRESENTATIVES RELATING to the PROJECTED INFLATIONARY IMPACT of the IMPLEMENTATION of the INFRASTRUCTURE INVESTMENT and JOBS ACT, the BUILD BACK BETTER ACT, and the INFRASTRUCTURE and JOBS ACT in CONJUNCTION with the BUILD BACK BETTER ACT. 2022. Congress.gov. https://www.congress.gov/congressional-report/117th-congress/house-report/588/1?s=1&r=8.

[41] RESOLUTION of INQUIRY DIRECTING the SECRETARY of the TREASURY to TRANSMIT CERTAIN DOCUMENTS to the HOUSE of REPRESENTATIVES RELATING to the PROJECTED INFLATIONARY IMPACT of the IMPLEMENTATION of the INFRASTRUCTURE INVESTMENT and JOBS ACT, the BUILD BACK BETTER ACT, and the INFRASTRUCTURE and JOBS ACT in CONJUNCTION with the BUILD BACK BETTER ACT. 2022. Congress.gov. https://www.congress.gov/congressional-report/117th-congress/house-report/588/1?s=1&r=8.

[42] Strand, Mark. 2013. "The Hastert Rule." Congressional Institute. July 17, 2013. https://www.congressionalinstitute.org/2013/07/17/the-hastert-rule.

The appropriation of funds reached federal departments such as the EPA after the passage of the IRA, thus beginning the implementation process. An interview with Peter Hansel, a consultant for the EPA leading the Climate Pollution Reduction Grant, revealed insights into this process. The EPA's role in implementing the IRA has been to assess where the money appropriated will decide how to spend billions responsibly to accomplish these goals according to the required statutes in a way that has a good impact. They also have to grow the infrastructure within the EPA to do so. With this unique grant program, there was a lesser level of opposition, primarily when awarded to many communities across the United States that recognize the economic value of clean energy.

Partisanship allowed for the survival of this law, as with majorities in all bodies of Congress, party members voted and acted in line with each other and perpetuated the Biden Administration's Build Back Better Agenda. Numerous amendments with considerable support were near passage that would have changed the Inflation Reduction Act's outcome to one that could have put the United States behind its climate goals. Using the congressional power of appropriations instead of Command-and-Control legislation based on the Commerce Clause,[43] the Inflation Reduction Act has reduced legal vulnerabilities that the Supreme Court could exploit.[44] The extreme polarization on the federal level does not follow suit in the local and state government levels. Peter Hansel from the EPA revealed that 46 states across diverse political spectrums have applied to receive this money, showing bipartisan enthusiasm at the state level. They are eager to receive some of the 4.6 billion set aside in implementation grants. Local metropolitan areas, like Raleigh-Cary, have collaborated to apply for funding, which received 1 million to develop their plan. These joint funding applications and connected sustainability planning are new, displaying the need for collaboration in climate change, contrary to extreme partisanship on the federal level. Encouraging emissions reductions with funding and tax credits and the debates on the effects of the Inflation Reduction Act on inflation have profound economic implications.

Economic Implications of the Inflation Reduction Act

Economic Foundations

While the IRA was protected through the legislative process by partisan lines, its implementation is primarily economic. Even its little-known sticks, such as the methane fee, are mainly an economic attack on producers of the greenhouse gas.[45] Though this doesn't mean that financial forms of mitigating climate change are ineffective. On the contrary, numerous forms of economic policies could drive climate policy. This can be in the form of subsidies that stimulate domestic clean energy manufacturing. Giving subsidies to green projects doesn't always float so well within the United States Congress, which is why there is a unique part of the IRA that gives it its name. The bill isn't only budget neutral, but even budget negative. This is shown by the IRA spending an estimated $485 billion on various initiatives to counter climate change and some funding to continue a couple of the Obama Administration-era *Affordable Care Act* subsidies, it is actually cutting the budget by between $238 billion-$305 billion through various methods.[46] As mentioned in the previous section, this was an entirely bipartisan bill even though cutting spending towards the national budget is generally seen as a republican standpoint.

[43] Khan Academy. 2019. "Command-And-Control Regulation (Article) | Khan Academy." Khan Academy. 2019.
[44] Lazarus, Richard, and Sara Zdeb. 2021. "Environmental Law & Politics." Www.americanbar.org. January 5, 2021.
[45] Jonathan Ramseur, "Inflation Reduction Act Methane Emissions Charge: In Brief - CRS Reports," Congressional Research Service, August 29, 2022 pp. 2.
[46] "What's in the Inflation Reduction Act?," Committee for a Responsible Federal Budget, July 28, 2022.

Nonetheless, the IRA's subsidies will help facilitate the development of multiple green industries and jumpstart America's development in the international race to lower their nation's impact on climate change. The Inflation Reduction Act is an exclusively economic-based means of stimulating climate reduction industries through a barrage of various subsidies and tax credits. The act's impacts are not only to stimulate industries but to change them altogether. These subsidies will: increase domestic production of green industries, create more green jobs that will be more viable in the future while also protecting communities that rely on fossil fuels, reducing the cost of living within aspects of American's life such as energy and healthcare, and do all of this while not increasing the federal budget a single dollar but reducing it by billions.

Subsidy Distribution

The Inflation Reduction Act is primarily a subsidized piece of legislation to help facilitate the energy revolution across the United States. Before getting into the impacts of the piece of legislation, let's see where the budget of the bill is going and how much towards each initiative. The reason for this priority is due to how much the IRA has in its budget, which is $485 billion. To put this into perspective, the GDP of Nigeria, the largest economy by GDP in Africa, is only 477.39 billion.[47] This is an astronomical amount of money and only exemplifies its importance in developing green industries within the United States. To start, around 98 billion dollars of the IRA's budget will be given to health care initiatives.[48] Those initiatives include an extension of Affordable Care Act subsidies, a Low-income subsidy program (LIS), vaccine coverage, and a redesign of part-D within Medicare. Starting with the ACA subsidies, these will extend the previous enhanced subsidies due to expire in 2022 by three years. If these subsidies did expire premium payments in 2022 would have been 53% higher, around $700 on average if not for these enhanced subsidies.[49] The inclusion of these was to counter the impact of inflation as the rising price of Medicare was affecting "more than 40 million Americans who have coverage under the ACA"[50]. The programs for Medicare also include an increase of the LIS program to include anyone whose income is within 150% of the federal poverty level and the inclusion of vaccine coverage for adults for lethal ailments.[51] [52] As for the part-D redesign for Medicare, that will examined in the inflation portion of the study due to the impacts it had on lowering the federal budget. These initiatives work with other Biden-era administration bills, such as the American Rescue Plan, which also helps lower healthcare prices for Americans. Though these initiatives do cause a dent within the budget of the IRA, other health legislature present within the act will fund these programs. This will be more diligently examined during the actual inflation reduction part of the study. Still, an example of this is the Medicare inflation rebates which requires companies that raises the price of drugs above the rate of inflation to be forced to pay a rebate.[53] This cap alone brings $101 billion to the federal budget.[54]

These medical programs are vital, but make up only around one-fifth of the IRA budget. The core of the act is for energy and climate which holds a total of $386 billion of the budget.[55] Over forty percent of that budget will go towards clean energy tax credits, which will help with the creation of clean energy industries.[56] Most of that will be facilitated by the Investment Tax Credit (ITC) and the Production Tax Credit (PTC). Both of these credits are due to expire by

[47] "GDP (Current US$) - Nigeria," World Bank Open Data, accessed November 12, 2023.
[48] What's in the Inflation Reduction Act?," Committee for a Responsible Federal Budget, July 28, 2022,
[49] Krutika Amin, Cynthia Cox, and Jared Ortaliza, "Five Things to Know about the Renewal of Extra Affordable Care Act Subsidies in the Inflation Reduction Act," KFF, August 11, 2022.
[50] Rose Chu, "Health Coverage under the Affordable Care Act: Current Enrollment Trends and State Estimates," ASPE, March 23, 2023.
[51] "Updated: November 2, 2023 What Is the Ira Lis Provision?," CMS, accessed November 13, 2023,pp. 1.
[52] "Fact Sheet Anniversary of the Inflation Reduction Act: UPDATE on CMS Implementation," CMS.gov Centers for Medicare & Medicaid Services, accessed November 12, 2023.
[53] "Fact Sheet Anniversary of the Inflation Reduction Act: UPDATE on CMS Implementation," CMS.gov Centers for Medicare & Medicaid Services, accessed November 12, 2023,
[54] What's in the Inflation Reduction Act?," Committee for a Responsible Federal Budget, July 28, 2022,
[55] What's in the Inflation Reduction Act?," Committee for a Responsible Federal Budget, July 28, 2022,
[56] What's in the Inflation Reduction Act?," Committee for a Responsible Federal Budget, July 28, 2022,

2036.[57] These credits aim to create and expand previous tax incentives to create more green energy projects. The ITC takes up around 50.858 billion dollars from the budget and provides a six percent tax credit for qualified investment.[58] [59] As for the PTC, it provides "0.3 cents/kW, inflation-adjusted" credit that comes from the act's budget of $11.204 billion with an extension of previous credits adding another $51.062 billion.[60] [61] While these initial credits may seem non-influential due to their low level, multiple bonus credits apply to them if the person producing the power or investing in green energy programs is eligible. These bonus credits consist of the following: Labor requirements which seek producers to meet wage and apprenticeship requirements, Domestic Content Bonus which calls for all iron and steel and at least 40% initially of the manufacturing to occur within the United States; Energy Community Bonus, and for the ITC a low-income bonus.[62] [63] The requirements for the domestic content bonus do increase over time to a cap of 55% but provide investors and producers with a 10% increase in the tax credit for both the ITC and PTC.[64] This provides an incentive for producers to increase their savings on their projects and allows the United States to slowly become more self-sufficient in its production of renewable energies and control over its production of steel and iron which was previously overseas. Though the benefit for the domestic industry doesn't end there, the energy community bonus gives another 10% bonus if the investor or producer has their project within a community that was previously being held up by the fossil fuels industry.[65] This allows communities that would otherwise be devastated by the energy transition to have the ability to continue to thrive within the energy sector but under the new guise of green energy. The low-income bonus also can help with this as coal-producing regions are significantly more prone to poverty, especially in the case of Appalachia.[66] The ITC and PTC are mutually exclusive, meaning that if a person wishes to make an energy product, they are only allowed to choose one of the presented credits, and not both. This allows for more projects to be able to gain access to the benefits without some developers hoarding the credit for themselves. However, there are different benefits to choosing each. If a certain project has a high investment cost, then it would be more beneficial to use the ITC. On the other hand, if the project can produce a substantial amount of clean energy, then the PTC would be a more viable option. The last prominent part of the energy and climate budget is the $30 billion Zero-Emission Nuclear Power Production Credit, which provides around $3-15/MWh depending on whether they meet the previous labor requirements.[67] This can make nuclear one of the most competitive forms of energy as "The average US nuclear plant cash cost is $29/MWh in 2021," thus bringing that cost down by the potential of $15/MWh through this initiative to around $10/MWh by 2024.[68] This could show the future potential of nuclear reactors as a primary source of energy creation within the United States and keep the industry competitive as other forms of green energy drop in pricing. However, the stigma of previous disasters must be overcome before true implementation can occur. Clean Energy isn't the only thing subsidized in the IRA. Air pollution and transportation have been allocated around $42 billion in funds and grants.[69] This comes primarily from the Greenhouse Gas Reduction Fund which provides $27 billion in competitive grants and is subdivided into "the $14 billion National Clean Investment Fund, the $6 billion Clean Communities Investment Accelerator, and the $7 billion Solar for All competition".[70] Most

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57 "Federal Solar Tax Credits for Businesses," Energy.gov, accessed November 12, 2023.
58 Congressional Budget Office revised August 5, 2022 cost estimate, accessed November 13, 2023,
59 "Federal Solar Tax Credits for Businesses," Energy.gov, accessed November 12, 2023,
60 Congressional Budget Office revised August 5, 2022 cost estimate, accessed November 13, 2023,
61 "Clean Energy Tax Provisions in the Inflation Reduction Act," The White House, September 21, 2023,
62 "Federal Solar Tax Credits for Businesses," Energy.gov, accessed November 12, 2023,
63 Keith Martin, David Burton , and Hilary Lefko, "Domestic Content Bonus Credit: Norton Rose Fulbright," Project Finance, accessed November 13, 2023,
64 Keith Martin, David Burton , and Hilary Lefko, "Domestic Content Bonus Credit: Norton Rose Fulbright," Project Finance, accessed November 13, 2023,
65 "Clean Energy Tax Provisions in the Inflation Reduction Act," The White House, September 21, 2023,
66 Eric Bowen, John Deskins, and Brian Lego, An overview of coal and the economy in Appalachia, 2020, p 33.
67 "Breaking down Clean Energy Funding in the Inflation Reduction Act," Motive Power, February 23, 2023,
68 "Ira Ripple Effect: 10 Areas of Impact," IRA ripple effect: 10 areas of impact, August 1, 2023, pp. 4
69 "Breaking down Clean Energy Funding in the Inflation Reduction Act," Motive Power, February 23, 2023,
70 About the Greenhouse Gas Reduction Fund | US EPA," United States Environmental Protection Agency, July 14, 2023.

notable of the three is the National Clean Investment Fund which provides capital to green banks and non-profit institutions. The purpose of this is not only to help major developers develop green energy projects but mostly for communities to have the capital for clean technologies. This is shown by the fact that “40% of capital is flowing into low-income and disadvantaged communities”.[71]

Though the bill doesn’t end there, there are one hundred and nineteen different initiatives that are a part of the IRA. These other initiatives cover diverse areas from consumer incentives to conservation. The remaining noteworthy incentives to discuss are the Advanced Manufacturing Production Credit which has $31 billion and the Clean Hydrogen Tax Credit which has $13.2 billion, and the Clean Vehicle Credit which has $7.5 billion.[72] [73] Similar to the rest of the subsidies and tax credits, the purpose of these three credits is to incentivize development within their respective sectors. Starting with the Advanced Manufacturing Production Credit, this is supposed to subsidize the industrial part of the energy revolution with tax benefits on specific critical components such as PV modules or batteries. This is shown by the credit of providing $12 per square meter of photovoltaic wafers produced within the United States, which is essential for the creation of solar panels.[74] This credit hits many underdeveloped parts of the United States economy concerning green energy and will help the industry not only develop but be more economically viable in the future. As for green hydrogen, the tax credit for that is making the industry viable by essentially cutting the price in half.[75] The reason for this is that the tax credit provides at most $3 per kg produced depending on the level of CO2 emitted.[76] The benefit of hydrogen doesn’t stop there as previous credits such as the PTC can be applied to hydrogen projects making the cost potentially more inexpensive. This is one of the key aspects of the IRA, its multidimensional coverage and overlapping subsidies. Especially with the inclusion of the previously mentioned PTC and ITC, no matter what is being generated in the field of clean energy there are numerous different incentives to not only create power generation projects but the potential to build entire supply lines for green energy. This along with the Biden Administration’s goal of jumpstarting the United States industry within green energy and related subfields are the core intentions of the IRA.

Job Creation

The creation of industries off of subsidies usually leads to one thing in particular, the creation of jobs. John Maynard Keynes once said, “If the Treasury were to fill old bottles with banknotes, bury them... and leave it to private enterprise …to dig the notes up again there need be no more unemployment”.[77] This statement still rings valid with the energy transition. According to a UMass Amherst study, “each $1 million shifted from brown to green energy will create a net increase of 5 jobs”.[78] The notable point in that statement is that there will be a net increase in the transition from fossil fuels to green energy. This simple fact causes sharp divides within the United States as some see the IRA’s potential to bring in around 9,121,587 jobs while others see how fossil fuel-based communities will be devastated by the change and lose around 900,000 and 1.2 million jobs by 2035.[79] [80] There has to be a loss towards the transition of anything, no matter if it’s as small as the loss of VHS to the DVD or as the energy transition

[71] 1. “National Clean Investment Fund | US EPA,” United States Enviromental Protection Agency, August 29, 2023.
[72] “Inflation Reduction Act Guidebook,” The White House, September 21, 2023.
[73] “Breaking down Clean Energy Funding in the Inflation Reduction Act,” Motive Power, February 23, 2023,
[74] Viktoria Vozarova, David Burton, and Jenna Goldman, “Inflation Reduction Act of 2022 – New Tax Credits for Manufacturers of Clean Energy Equipment: Norton Rose Fulbright,” Project Finance, August 11, 2022.
[75] Anshuman Saini, “Green & Blue Hydrogen: Current Levelized Cost of Production & Outlook: GEP Blog,” Green & Blue Hydrogen: Current Levelized Cost of Production & Outlook | GEP Blog, January 12, 2023.
[76] “Financial Incentives for Hydrogen and Fuel Cell Projects,” Energy.gov, accessed November 13, 2023.
[77] “John Maynard Keynes Quote,” accessed November 13, 2023,
[78] Heidi Garrett-Peltier, “Green versus Brown: Comparing the Employment Impacts of Energy Efficiency, Renewable Energy, and Fossil Fuels Using an Input-Output Model,” *Economic Modelling* 61 (2017): 439–47,.
[79] Chirag Lala, Robert Pollin, and Shouvik Chakraborty, “Job Creation Estimates through Proposed Inflation Reduction Act,” PERI, August 4, 2022, pp. 3.
[80] Devashree Saha, Rajat Shrestha, and Phil Jordan, “How a Clean Energy Economy Can Create Millions of Jobs in the US,” World Resources Institute, September 14, 2022.

from fossil fuels to clean energy. Though the change is inevitable, that doesn't mean that the communities that depend on nonrenewable energy should be left behind. The IRA agrees with that statement and backs it up with the fact that there are incentives to move into these communities and replace the fossil fuel industry with a low-carbon industry. This is shown with the previously mentioned energy community bonus which can be applied to multiple tax credits such as the ITC and PTC if the project in question is located within a community that is losing its fossil fuel jobs. This incentive is even working because "Republican-voting states, which have larger energy workforces…would lose around 700,000 fossil-fuel jobs but gain 1.7m low-carbon energy jobs between 2025 and 2050".[81]The forethought of including these communities in the transition not only increases the likelihood they survive and can thrive in the future but also allows future bills to potentially have more bipartisan support as Republican states realize the economic potential that future subsidies will have in creating new jobs within their state. This has already slowly started to materialize as Peter Hansel, an official from the EPA, states that 46 states have already requested funds from the Climate Pollution Reduction Grant.[82] This comes as no surprise as the 4.6 billion set aside for implementation funds would be tempting to any policymaker.[83] The energy transition cannot be done with one region alone. It must be a nationwide initiative that allows all communities to thrive and not allow resentment toward a greener future to block further development. The IRA will slowly achieve this with its initiatives to include both low-income communities as well as energy communities in the energy transition.

Inflation And Cost Of Living Reduction

The final critical aspect of the Inflation Reduction Act is whether it truly holds up to its name. This is primarily seen with the money saved by the average American and the money saved by the fact that the act reduces deficit spending. Starting with the latter, the IRA is cutting the federal budget by "$238 billion-$305 billion". The reason for this viable number is due to how it is primarily saving money, which is through the IRS or Internal Revenue Service. Within the IRA is a budget of $80 billion earmarked for closing up of loopholes in the tax code, thus providing more taxes to be collected, resulting in an estimated $204 billion in additional revenue.[84] This, along with a 15% minimum corporate tax, provides the IRA with $437 billion, which is nearly enough to cover the entirety of the bill itself but additional policy changes within healthcare allow for an additional $322 billion.[85] These health changes are primarily that of repealing costly policies from the previous administration. However, these changes also give the Centers for Medicare & Medicaid Services the ability to negotiate with pharmacies for the reduction of prices for certain pharmaceuticals and penalize those same pharmacies if they increase the price of their drugs above the rate of inflation.[86] Through these policies there isn't only a decrease in the budget but also the cost that Americans have to pay. In the case of Medicare, the ability to negotiate the cost of the most expensive drugs and the subsidies that follow the ACA will lower the cost of health insurance by $800 a year and the cost of insulin by $866 per year for Americans.[87] This along with policies such as medical inflation rebates and a price cap on insulin at $35 per month will lower the cost of healthcare. The savings don't stop there; the multiple subsidies and tax benefits towards energy will allow for cheaper energy over time as the act's effects begin to occur. This is shown by the fact that American families will be

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[81] Josh Gabbatiss, "Jobs Created by Net-Zero Transition Will 'offset' Fossil-Fuel Job Losses in Republican Us States," Carbon Brief, May 31, 2023.
[82]Hansel, Interview
[83]Hansel, Interview
[84] "What's in the Inflation Reduction Act?," Committee for a Responsible Federal Budget, July 28, 2022,
[85] "What's in the Inflation Reduction Act?," Committee for a Responsible Federal Budget, July 28, 2022,
[86] "Inflation Reduction Act and Medicare," CMS.gov, September 12, 2023,.
[87] "Fact Sheet: One Year in, President Biden's Inflation Reduction Act Is Driving Historic Climate Action and Investing in America to Create Good Paying Jobs and Reduce Costs," The White House, August 16, 2023.

able to save "$27-38 billion on their electricity bills from 2022-2030".[88] This isn't just due to the overall cheapening of electricity with the bill but also the tax incentives that provide families the funding to install cost-saving appliances, such as energy-efficient windows and the purchase of EVs, which will save money for them in the long term. Finally, reducing the federal budget allows for a corresponding reduction of inflation due to the Federal Reserve not having to print more currency to keep up with the national budget, thus limiting the amount of money within the circulation. However, this type of inflation reduction can be hindered if current policy changes reduce the impact of the reduction of the budget. This has already happened with the fact that Biden negotiated a cut of the IRS's budget by $14.3 billion to pay for emergency aid to Israel.[89] Policies like this don't take into account the amount of money lost after the fact since the IRS will become less capable of gaining revenue and closing the previously stated loopholes, thus weakening the impact of the IRA. Nor do they take into account the fact that the more bills like the IRA are stripped of their initial funding the more their influence on lowering the cost of living for Americans is decreased as funding is stripped from other potentially beneficial initiatives. Overall, as it currently stands the Inflation Reduction Act has numerous impacts on lowering the cost of inflation by lowering the prices of things like healthcare and energy for American people, and thus keeping true to its name.

The Unsteady Future Of The IRA

While being an economic piece of legislation that was exclusively supported on partisan lines, the Inflation Reduction Act will help develop the United States green industry in both blue and red states leading to cost of living savings and jobs for every American. This statement by itself has the prospect of making the idea of the energy transition not a polarizing topic but rather a bipartisan issue of job creation and energy self-sufficiency. However, until the effects of the IRA and climate bills similar to it take their place in the coming decades it will be up to open-minded politicians to keep the integrity and strength of the bill in action. This is a task for both sides of the political spectrum with the previously mentioned deal made by Biden to get emergency aid to Israel thus increasing the financial burden on taxpayers and multiple Republican presidential candidates have already promised the scraping of the IRA if they are elected such as former-president Donald Trump.[90] [91] Nonetheless, while the continuation and safeguarding of the IRA are vital to jumpstart the green energy transition within the U.S., future policies and legislation must be passed to keep up its momentum towards a more environmentally sustainable and equal future.

<u>Evaluating the Impact of the Inflation Reduction Act on American Households</u>

IRA Attempts to Expand Healthcare Affordability

The IRA contains several provisions to reduce healthcare costs for American households. Some of the most vital points of the IRA are allowing Medicare to negotiate prescription drug prices, restructuring Medicare Part D to cap out-of-pocket (OOP) drug costs, and expanding Affordable Care Act (ACA) insurance subsidies[92]

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[88] "Fact Sheet: One Year in, President Biden's Inflation Reduction Act Is Driving Historic Climate Action and Investing in America to Create Good Paying Jobs and Reduce Costs," The White House, August 16, 2023
[89] Katie Lobosco, "Why the GOP Plan to Cut IRS Funds to Pay for Israel Aid Would Increase the Deficit | CNN Politics," CNN, November 2, 2023.
[90] Alexander Rifaat, "I'll Scrap Ira Tax Credits on Day 1, Trump Says: Tax Notes," Tax News, Tax Articles and Information - Tax Notes, September 29, 2023,
[91] Kevin Breuninger, "Tim Scott Praises the 'laffer Curve' as He Vows to Make Trump-Era Tax Cuts Permanent," CNBC, September 14, 2023,
[92] Gellad, Walid F., and Inmaculada Hernandez. 2022. "Pharmaceutical Spending in Fee-For-Service Medicare." *JAMA* 328 (15): 1502.

Negotiating Medicare drug prices will lead to patient savings amounting to an expected $2 billion increase in government Medicare spending[93]. Still, the scope is initially limited, applying only to some high-cost drugs that have been around for a while. It is reasonable to be concerned that working citizens covered under self-paid insurance plans will face higher prescription costs as drug companies look to offset the losses of lower drug prices under Medicare. So far, there are no provisions in place to prevent drug companies from preying on self-insured or uninsured patients, which could lead to significantly higher launch prices for new drugs.

Medicare Part D covers prescription drug costs, and restructuring Part D Medicare provides impactful savings through the 2025 $2,000 OOP spending cap, which modeling predicts will collectively save beneficiaries over $1.7 billion annually[94]. In 2024, the cap will be ~$3,250 for Medicare beneficiaries. The caveat, however, is that in 2018, less than 14% of employer-insured persons reported spending $1,000 or more on drug costs[95]. Unfortunately, the IRA's Medicare reforms do little to address affordability challenges for those covered by private insurance plans. Medicare is intended for disabled persons and those above 65 years old. According to the US Federal Reserve, about 65% of Americans are covered by private insurance. Although the ACA subsidies have been extended for three more years under the IRA, the benefits of the IRA's proposal are intangible for most American citizens.

While the IRA's healthcare measures promote important cost-reduction goals, policymakers must expand the program to broaden the swath of patients across all insurance types to tackle affordability gaps. The IRA also contains no equity provisions, which leaves concern for racial disparities, as minority groups face higher rates of chronic illness and financial barriers to care that may persist without improved legislation[96].

Uncertain Impact on Household Energy Costs

The IRA's massive clean energy investments and improving household affordability could lower household energy costs, but uncertainties persist in this sector as well. Tax incentives for "efficiency upgrades" and rooftop solar may expand access to cost-saving measures[97]. According to the Solar Energy Industries Association, "An average-sized residential system has dropped from a pre-incentive price of $40,000 in 2010 to roughly $25,000 today…"[98] For most American households, $25,000 is far from affordable to switch to clean energy. These kinds of tax incentives covered by the IRA are expected to benefit higher-income households, which bypasses the intent of the IRA to provide relief for low-income households suffering from living costs in the US.

Over time, some tax benefits may fall within reach of more households thanks to R&D initiatives related to clean energy and agriculture. The Advanced Manufacturing Production tax credit incentivizes domestic production of batteries, solar panels, and other components essential for clean-energy products. This leads to lower costs for production and lower costs for purchasing these products, including solar energy panels and electric vehicles[99]. Clean energy grants could increase affordable access to efficiency improvements for select groups like public housing residents. The IRA's climate measures could provide substantial public health benefits,

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[93] Nasir, Khurram, Joseph Salami, and Elias Mossialos. 2023. "Prescription for Affordable Health Care." *National Library of Medicine* 81 (15): 1502–4.

[94] Zheng, Jimmy, and Alexander T. Sandhu. 2023. "The Inflation Reduction Act Expands Prescription Drug Affordability." *Journal of the American College of Cardiology* 81 (21): 2112–14.

[95] National Cancer Institute. 2020. "Cancer Statistics." National Cancer Institute. Cancer.gov. September 25, 2020.

[96] Nasir, Khurram, Joseph Salami, and Elias Mossialos. 2023. "Prescription for Affordable Health Care." *National Library of Medicine* 81 (15): 1502–4. https://doi.org/10.1016/j.jacc.2023.02.028.

[97] Levy, Jonathan I. 2022. "2022 Inflation Reduction Act: Climate Investments Are Public Health Investments." *American Journal of Public Health* 112 (11): 1525–25. https://doi.org/10.2105/ajph.2022.307089.

[98] SEIA. 2023. "Solar Industry Research Data | SEIA." SEIA. 2023. https://www.seia.org/solar-industry-research-data.

[99] DiGangi, Diana. 2023. "Treasury Issues Further Guidance for IRA's $10B Tax Credit Incentivizing Clean Energy Manufacturing." Utility Dive. June 1, 2023.

but their impact on immediate energy cost relief needs to be expanded for greater clarity and political longevity.

Impact On The US Labor Market

The IRA is expected to impact the US Labor Market substantially, making it both economically appealing and environmentally sustainable. According to figures from the political research institute of UMass Amherst, the provisions covered under the IRA are expected to create more than nine million new job positions over the course of ten years[100].

TOTAL JOB CREATION ESTIMATES
Summary Figures

	Average Annual Budget and Job Creation Figures over 10 Years			Total Budget and Job-Years Figures	
	Public Spending	Total Spending *(= public + private spending)*	Annual Job Creation	Total Spending	Total Job Creation, Job-Years
Electricity Programs	$21.4 billion	$66.3 billion	573,177	$663 billion	5,731,771
Transportation Programs	$3.3 billion	$5.6 billion	65,821	$56 billion	658,212
Building Programs	$4.8 billion	$9.4 billion	91,082	$94 billion	910,819
Manufacturing Programs	$6.7 billion	$12.8 billion	106,032	$127.7 billion	1,060,320
Environmental Justice and Community Resilience Programs	$1.1 billion	$1.1 billion	14,892	$11 billion	148,916
Lands Programs	$1.1 billion	$1.1 billion	22,582	$10.9 billion	225,817
Agriculture Programs	$2.2 billion	$2.2 billion	38,573	$22 billion	385,732
TOTALS	$40.6 billion	$98.4 billion	912,159	$984 billion	9,121,587

Figure 1. Chirag Lala, Robert Pollin, and Shouvik Chakraborty, "Job Creation Estimates through Proposed Inflation Reduction Act," PERI, August 4, 2022, pp. 3.

This new job creation would boost economic productivity and provide greater job security and incentivize people working in non-renewable industries such as coal. Its provisions include benefits such as wage replacement and retirement, not to mention the substantial health benefits of leaving such sectors. As mentioned in our interview with Jennifer Weiss, this boom in job creation, paired with potentially reduced spending on essential living costs, could lead to

100 Weinstein, Kim. 2021. "PERI - the Economic Effects of Investing in Quality Care Jobs and Paid Family and Medical Leave." Peri.umass.edu. June 14, 2021.

more disposable income, boosting the market again. The IRA also provides funding for training programs to prepare the current workforce for said transitions.

Future Potential

The Inflation Reduction Act delivers significant investments across healthcare, energy, taxes, and debt relief, potentially providing American households with considerable cost savings and economic opportunities. However, ambiguities persist regarding the real-world impacts of these reforms. Key healthcare provisions may only benefit certain groups like Medicare beneficiaries. Without additional subsidies, clean energy incentives could have limited effects on low-income families' energy costs. Credit and debt relief measures leave out many vulnerable groups but could still lead to positive externalities for all communities. The IRA's goals of broadly enacted financial relief and equity require targeted reforms that will address limitations. Continued tracking of effects on affordability across demographics will also be necessary to understand the fuller scope of the IRA's impact. With diligent policy adjustment and monitoring, the IRA's groundbreaking investments can live up to their promise and potential.

Evaluation of the Environmental Implications of the IRA

The Inflation Reduction Act represents a significant legislative effort to tackle the environmental and climate consequences of harmful activities, primarily by allocating funds to facilitate the energy transition. As concerns over the planet's well-being escalate, policymakers have grappled with the imperative to curb ecological damage while addressing economic considerations. The Act, with its focus on mitigating inflation and concurrently addressing environmental issues, marks a commendable step forward. Examining the Act's merits, it can be observed that there is positive impact on environmental initiatives but not without limitations. The importance of this issue is underscored by the pressing need for a sustainable future; the delicate balance between economic stability and environmental conservation necessitates a thorough evaluation of the Act's provisions. While the legislation takes laudable strides, there exists a discernible gap between its aspirations and the urgency of the environmental crisis. This analysis delves into the Act's provisions, evaluates arguments from multiple perspectives, and seeks to draw nuanced conclusions. It is imperative to not only recognize the Act's positive contributions but also address its shortcomings. We must navigate through the complexities, acknowledging opposing viewpoints, and ultimately offering recommendations to enhance the Act's efficacy in fostering a more sustainable and resilient future.

Environmental Impact of Energy Systems

The environmental impact of energy systems, particularly the shift from coal to renewable sources, is a pivotal consideration in evaluating the efficacy of the Inflation Reduction Act. Coal, a historically dominant energy source, is associated with deleterious environmental effects, as illuminated by various studies. Coal mining causes pervasive harm, demonstrated in a study of a supposedly protected Australian river, which experienced critical water pollution and ecological impairment[101]. Furthermore, significant household air pollution has been linked to coal combustion in China[102]. The harmful impacts of coal are multidimensional, encompassing air pollution, water pollution, and significant contributions to carbon emissions and global warming. Conversely, adopting renewable energy systems constitutes a pivotal aspect of the Act's agenda. Reducing coal burning, a principal tenet of the energy transition, is crucial in mitigating these negative repercussions. It has been established that global methane emissions from coal mining persist despite declining coal production[103], underscoring the persistent environmental challenges associated with traditional energy sources.

Renewable systems offer a promising alternative by concurrently addressing air pollution and carbon emissions[104]. The Act's commitment to funding the energy transition aligns with the broader global imperative to mitigate the harmful environmental impacts linked to conventional energy sources. It is important to explore the extent to which its provisions effectively navigate the complexities of this transition and acknowledge potential shortcomings in realizing a truly sustainable energy landscape.

IRA’s Effect on Emissions

The IRA sets an ambitious course for the United States by aiming to double the annual emissions reduction rate from 2% to 4%. This strategic escalation is underscored by findings from Bistline et al. (2023), emphasizing the pressing need to address the environmental consequences of emissions and position the nation on a sustainable trajectory.

Historically, the U.S. has been making commendable progress with a 2% annual reduction in emissions. However, the IRA heralds a pivotal shift in ambition by seeking to amplify this rate. Bistline et al.'s (2023) study on the emissions and energy impacts of the IRA provides a foundational understanding of the potential efficacy of this legislative initiative. The envisioned doubling of the annual reduction rate is intrinsically tied to the IRA's commitment to funding the energy transition. By directing resources toward renewable energy systems, the Act endeavors to reduce reliance on environmentally detrimental energy sources and stimulate the adoption of cleaner alternatives. This strategic investment aligns with the global imperative to combat climate change and underscores the IRA's proactive approach to mitigating the environmental impact of harmful activities. However, a comprehensive analysis must grapple with the intricate interplay of economic, technological, and geopolitical factors that could influence the actualization of the proposed reduction rate. Additionally, it is imperative to scrutinize the IRA's capacity to address sector-specific emissions differentials and its resilience in the face of potential opposition or skepticism. In doing so, a nuanced evaluation will highlight the Act's potential to significantly contribute to reducing U.S. emissions and its broader role in fostering a sustainable environmental future.

[101]Wright, I. A., Belmer, N., & Davies, P. J. (2017). Coal Mine Water Pollution and Ecological Impairment of One of Australia’s Most ‘Protected’ High Conservation-Value Rivers. Water, Air, and Soil Pollution, 228. https://doi.org/10.1007/s11270-017-3278-8
[102]Zhang, J., & Smith, K. R. (2007). Household Air Pollution from Coal and Biomass Fuels in China: Measurements, Health Impacts, and Interventions. Environmental Health Perspectives, 115(6), 848-855. https://doi.org/10.1289/ehp.9479
[103]Kholod, N., Evans, M., Pilcher, R. C., Roshchanka, V., Ruiz, F., Coté, M., & Collings, R. (2020). Global methane emissions from coal mining to continue growing even with declining coal production. Journal of Cleaner Production, 256. https://doi.org/10.1289/ehp.9479
[104]National Academies Press. (2011). Environmental Impacts of Renewable Electricity Generation. In The Power of Renewables: Opportunities and Challenges for China and the United States (pp. 89–111). essay.

While the IRA signifies a commendable effort to intensify the reduction of U.S. emissions by doubling the annual rate to 4%, a critical evaluation reveals that this initiative needs to align with the imperative for net-zero emissions targets[105]. The assessment draws from insights a study on energy requirements and carbon emissions for a low-carbon energy transition offers valuable perspectives on the trajectory needed for substantial environmental impact. To meet the ambitious target of reducing emissions by 50% below peak levels by 2030 and achieving net-zero emissions by 2050, the U.S. must aim for a more aggressive annual reduction rate of 6%[106]. This necessitates recalibrating current efforts to further align with global environmental objectives. The findings of Slameršak et al. (2022) underscore the scale of the challenge, emphasizing that a low-carbon energy transition requires an increase in renewable energy adoption and a comprehensive restructuring of energy systems. While the IRA's commitment to a 4% annual reduction is a step in the right direction, it is crucial to acknowledge that achieving net-zero emissions demands more substantial and expeditious measures. A robust strategy should encompass technological innovations, stringent regulatory frameworks, and proactive collaboration with diverse stakeholders. It is essential to pursue a holistic and accelerated approach to meet the stringent net-zero emissions targets.

Where to Go From Here?

To conclude, the IRA is pivotal, acknowledging and tackling the environmental repercussions of harmful activities by doubling the annual U.S. emissions reduction rate to 4%. This legislation reflects an awareness of the pressing need to shift towards sustainable energy systems, addressing the far-reaching impacts of climate change. The nuanced scrutiny brings forth both positive facets and inherent limitations within the IRA. The Act's focus on financing the energy transition signals a strategic shift, showcasing a deliberate effort to pivot from environmentally harmful energy sources to cleaner alternatives[107]. This underscores a cognizance of the intricate balance between economic stability and environmental responsibility. However, viewed critically, the IRA's commendable commitment falls short of the more ambitious targets needed for achieving net-zero emissions.

As the analysis unfolds, it becomes apparent that additional measures must align with the stringent environmental benchmarks of a 50% reduction below peak levels by 2030 and net-zero emissions by 2050. Insights from Slameršak et al. underscore the necessity for recalibration, incorporating more ambitious reduction rates, embracing technological innovations, and facilitating a comprehensive restructuring of energy systems. Such recalibration is crucial for effectively navigating the complex challenges of the environmental crisis and fostering an impactful, sustainable response.

Environmental stewardship demands proactive and adaptive strategies. While indicative of progress, the IRA should be perceived as an initial stride rather than a definitive solution. It

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serves as a foundational step into the intricate terrain of achieving net-zero emissions. The lessons drawn from both the strengths and limitations of the IRA should shape subsequent policy initiatives, guiding the nation toward a more robust and effective response to the urgent environmental concerns of our time. As the U.S. continues its trajectory toward a sustainable future, an ongoing commitment to innovation, inclusivity, and adaptability remains paramount in shaping policies that genuinely address the multifaceted challenges of environmental preservation and economic stability.

Conclusion

The Inflation Reduction Act (IRA) represents a significant legislative achievement to tackle critical issues like climate change, healthcare affordability, and economic growth. Through nearly $500 billion in investments, incentives, and cost-saving measures, the IRA has the potential for wide-ranging impacts on American society.

However, the full effects of the IRA remain uncertain. As the case study shows, partisan politics and legislative compromises shaped the final bill, limiting its scope in some areas. Key provisions target select groups like Medicare beneficiaries for prescription drug savings, while private insurance holders may still struggle with rising healthcare costs. Clean energy tax credits could spur domestic manufacturing and deployments, but their ability to directly lower household energy bills is unclear absent further subsidies.

Ultimately, whether the IRA lives up to its promises relies heavily on implementation and monitoring. Achieving broadly shared financial relief and equity requires diligent tracking of impacts on affordability across demographics. Policy adjustments addressing remaining coverage and affordability gaps will be vital. The IRA’s investments are essential, but realizing its full potential depends on ongoing reforms to promote accessibility, equity, and effectiveness across all its major initiatives. Similarly, its emphasis on the electrification of end uses, such as hydrogen production, could pressure the grid, having a negative externality in increasing the need for fossil fuels to supplement energy generation (Interview with Peter Hansel).

In addition, it is not clear whether the effectiveness of the IRA will hold up if there is a change in party leadership for the coming elections. Partisanship protected this legislation in its passage, but House Republicans, as early as October of 2022, were calling for repeals of IRA provisions that helped pay for its services (Statement by President Biden’s Administration). Though the act might be dismantled slowly if its components are continuously stripped away to make compromises.

Furthermore, statements from prominent conservatives, as collected by the Democratic Party Website, display their intent to hinder the effects of the IRA. Sen. Tim Scott goes as far as to say he “would simply eliminate the IRA, the Inflation Reduction Act, which, of course, is a lie from the pit of hell” (Democratic Party Website). EPA departments that manage grant programs such as the Climate Pollution Reduction Grant have aggressive timelines in anticipation of this, with plans to disperse all funding by the end of December 2024. However, they have plans to continue oversight through 2027 at least (Interview with Peter Hansel). Peter Hansel from the EPA is relatively optimistic that, at the very least, tax incentives will continue regardless of near-future political changes since there is a lot of state and local support for those.

With careful stewardship, this legislation, the IRA can drive progress on its interlinked climate action goals, healthcare access, good jobs, and cost reductions for American families.

This potential warrants robust follow-through by policymakers to ensure the IRA's historic investments translate into real-world improvements for all.

Appendix: Interview Notes

An interview with Jennifer Weiss, Co-Director of the NC Clean Energy Fund, an emerging Green Bank in North Carolina, revealed an insight to the processes of actually applying for funding under the IRA. The NC Clean Energy Fund exists to fill market gaps, such as low income households or small businesses that don't have the credit to qualify for renewable upgrades. Green Banks are supported by the IRA through the Greenhouse Gas Reduction Fund, providing financing opportunities to accelerate their operations.

She notes how the IRA has elevated the conversation of climate change in North Carolina to be action-oriented, where partnerships are created around important issues like coastal resiliency, and these pots of money help implement projects.

In regards to inflation, she says that this law doesn't reduce inflation but reduces the burden on households."Whenever you do something to reduce energy burden and bills, that will improve household abilities to purchase other things, creating disposable income."

Since the passage of this law as well, in discussion with prominent conservative leaders on the NC State or local government levels, she has noticed that with the right framing, these leaders are open to renewable development since they create more jobs. "You have to talk about the right benefit with the right audience."

An unintended effect of the IRA is the fear that the money isn't being spent the right way. Jennifer Weiss's biggest worry is that the money isn't actually going to go to the 40% disadvantaged communities that were written into the original law. Another effect is that one funded program could become a political dartboard and that publicity could disrupt the other good. Smaller groups applying for funding from the IRA sometimes don't have the experience nor time to navigate the federal grant system, which is already not a streamlined process. These groups may not have the capacity to devote so much time to applying for the correct paperwork. Finally, on the nonprofit level, there is an awareness gap in such expansive legislation on what funds these organizations are eligible to apply for.

We conducted another interview with Peter Hansel, a consultant for Federal Climate and Energy Policy, currently working for the EPA leading the Climate Pollution Reduction Grant. He was hired by the EPA to implement the IRA, specifically with the Climate Pollution Reduction Grant that provides grants to state and local governments to put together greenhouse gas reduction plans and provide funds to the same entities to implement those plans. He says this is the first time the federal government has awarded money for climate planning at this scale (5 billion dollars, already awarded 250 million in planning grants). 46 states across diverse political spectrums have applied to receive this money, showing bipartisan enthusiasm at the state level. They are eager to receive some of the 4.6 billion set aside in implementation grants.

The EPA's role in implementing the IRA has been to assess where the money appropriated is going, to decide how to spend billions responsibly that accomplishes these goals according to the required statutes in a way that has good impact. They also have to grow the infrastructure within the EPA to do so. For potential opposition, the unique thing about grants programs is that it has a lesser level of opposition, especially when you award grants to many different communities across the United States, states and localities see the economic value in investing in clean energy.

They are moving on an aggressive schedule to get the planning grant money out by October or December 2024, though they will continue assessing the oversight of these funds -

with a timeline to develop an initial plan by March 2024 and follow with a more comprehensive plan in may-ish 2025 and a status update in 2027.

Local metropolitan areas have collaborated to apply for funding, like Raleigh-Cary which received 1 million to develop their plan. Sometimes these metro areas coordinate on some issues like transportation, but otherwise, they don't collaborate, making these joint funding applications and connected sustainability planning very new, displaying the need for collaboration in climate change.

Having the IRA as proof of the US' commitment to reducing emissions will help encourage international efforts like COP28. The more that the government can do between now and 2024 to demonstrate the value of the IRA, the more likely it will stand the test of time or political attacks. Peter Hansel is fairly optimistic that at the very least tax incentives will continue regardless of near-future political changes since there is a lot of support for those.

A personal thing Peter Hansel is curious about is the hydrogen tax incentive under the IRA. There are many ways to produce hydrogen and it is difficult to implement. Another thing that is important to track is electricity - if the electrification of end uses are intensified, will it increase demand for fossil fuel units that would have otherwise been phased out?

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